



11.0.1 ISRO-DEC2017-6



The proposition $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$ is a

- A. Tautology B. Contradiction C. Contingency D. Absurdity

isrodec2017 discrete-mathematics mathematical-logic easy

Answer key

11.1

First Order Logic (2)

11.1.1 First Order Logic: GATE CSE 2004 | Question: 23, ISRO2007-32



Identify the correct translation into logical notation of the following assertion.

Some boys in the class are taller than all the girls

Note: $\text{taller}(x, y)$ is true if x is taller than y .

- A. $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
 B. $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
 C. $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$
 D. $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$

gatecse-2004 mathematical-logic easy isro2007 first-order-logic

Answer key

11.1.2 First Order Logic: ISRO CSE 2020 | Question: 73



Given that

$B(a)$ means “ a is a bear”

$F(a)$ means “ a is a fish” and

$E(a, b)$ means “ a eats b ”

Then what is the best meaning of

$\forall x[F(x) \rightarrow \forall y(E(y, x) \rightarrow b(y))]$

- A. Every fish is eaten by some bear B. Bears eat only fish
 C. Every bear eats fish D. Only bears eat fish

isro-2020 mathematical-logic first-order-logic normal

Answer key

11.2

Propositional Logic (1)

11.2.1 Propositional Logic: ISRO CSE 2017 | Question: 22



Which one of the following Boolean expressions is NOT a tautology?

- A. $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$
 B. $(a \leftrightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$
 C. $(a \wedge b \wedge c) \rightarrow (c \vee a)$

D. $a \rightarrow (b \rightarrow a)$

isro2017 mathematical-logic propositional-logic

Answer key 

Answer Keys

11.0.1	C	11.1.1	D	11.1.2	D	11.2.1	B
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